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An Objective Method to Locate Shear Lines during the Northeast Monsoon Season in the Philippines

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Abstract

The shear line is a narrow zone of maximum horizontal wind shear, typically identified as a confluence zone of low-level wind streams. Previous studies showed that this system can trigger heavy to extreme rainfall events in the Philippines during the northeast monsoon season. However, a research gap remains in objectively identifying and locating this system. Thus, this study develops a detection method that may be used for monitoring and forecasting the location of shear lines. Results show that the gradient of meridional winds in the y-direction ($\frac{\partial V_{925hPa}}{\partial y}$), the boundary layer moisture flux convergence, and the 925 hPa relative vorticity may be used to estimate the location of the shear line. These parameters are then utilized in the detection algorithm. Applying the detection algorithm to reanalysis data from 1979 to 2022 for the northeast monsoon season revealed that shear lines contribute up to 20% of the total extreme rainfall days, with the greatest influence observed in November and December. This system is also best depicted in the 925 hPa and 850 hPa levels. The detection method developed in this study may be useful for improving the forecast of heavy to extreme rainfall events in the Philippines, especially during this season.

Keywords: shear line, confluence zone, col, gradient of meridional winds, heavy rainfall

1. Introduction

During the southwest monsoon season in the Philippines (May to September), tropical cyclones (TCs; Cinco et al. 2016; Bagtasa 2017; Basconcillo and Moon 2020; Petilla et al. 2025), low-pressure systems (LPS) that are weaker than TCs (Olaguera et al. 2023), the Boreal Summer Intra-

Seasonal Oscillation (BSISO; Olaguera et al. 2022), and enhanced southwest monsoon by TCs (Cayanan et al. 2011; Bathan et al. 2025) are reported to contribute to rainfall extremes in this season. For example, during the active and strong phases of the BSISO, the probability of extreme rainfall occurrence in the Philippines can increase to as much as 80% in some regions. The active phases of the BSISO can also facilitate the development of more TCs over the western North Pacific that will eventually affect the Philippines. On the other hand, systems such as LPS (Olaguera et al. 2023), the Madden Julian Oscillation (MJO; Pullen et al. 2015; Bagtasa et al. 2020; Olaguera et al. 2024a), cold surges (Lim et al. 2017; Abdillah et al. 2022), and shear lines (Olaguera and Matsumoto 2020; Olaguera et al. 2021a, b; Reyes et al. 2024) contribute to the rainfall extremes during the northeast monsoon season (November to March), particularly along the eastern coast of the Philippines. Olaguera et al. (2024) quantified the contribution of MJO to be as much as 16% of the rainfall extremes in some stations along the eastern coast of the country in this season. TCs also contribute to the extremes during the northeast monsoon season (e.g., Espinueva et al. 2012; Petilla et al. 2025), although they are less frequent compared to those during the southwest monsoon season (e.g., Cinco et al. 2016; Bagtasa 2017).

A shear line is a narrow zone of maximum horizontal wind shear, typically identified by a confluence of air streams on the equatorward side of a *col* (Ramage 1995; Pielke 2013). This zone is termed as a shear line because the winds across it may blow in opposite directions or display sharp differences in horizontal wind speed (Pielke 2013). A *col* represents a saddle-like pressure pattern, located between two low-pressure systems and two high-pressure systems (https://glossary.ametsoc.org/wiki/Col). Shear lines commonly form when mid-latitude cold fronts intrude into the tropics, where their thickness gradients weaken upon encountering the warm tropical oceans. Low-level convergence, aided by convective activity, may persist during the

conversion process of the cold front to a shear line (Yokoi and Matsumoto 2008; Pielke 2013). It can also interact with a low-level cyclonic vortex in the central and southern Philippines, which hinders its southward propagation (Olaguera et al. 2021; Reyes et al. 2024). Therefore, determining the location of the shear line is crucial for identifying areas that are at risk of heavy and extreme rainfall.

However, the impact and mechanism of this system have only been recently examined by a few studies (Faustino-Eslava et al. 2011; Olaguera and Matsumoto 2020; Olaguera et al. 2021a,b; Reyes et al. 2024) despite the catastrophic impact that they may bring. Reyes et al. (2024) recently examined the heavy rainfall events (HREs) during the northeast monsoon season in the Philippines. They identified three patterns of HREs associated with a strong East Asian Winter Monsoon and the interaction of a shear line with a low-level cyclonic vortex in the vicinity of the Philippines. However, there is no available index or method that can be used to locate the shear line objectively. Therefore, the purpose of this study is to develop a detection algorithm that may be used for monitoring the shear line. This detection method will also be useful for quantifying the contribution of shear lines to the rainfall extremes during the northeast monsoon season in the Philippines, for identifying areas that are more likely to experience heavy to extreme rainfall events, and for improving our understanding of how this system evolves. The rest of the paper is organized as follows. The data used in this study are briefly presented in Section 2. The development of the shear line detection algorithm is then discussed in Section 3. The climatological features of shear lines and their contribution to rainfall extremes in the Philippines are described in Section 4. Finally, summary and concluding remarks are presented in Section 5.

2. Data

The 6-hourly reanalysis data of zonal winds, meridional winds, specific humidity, surface pressure, temperature, and relative humidity at multiple pressure levels from the Japan Meteorological Agency (JMA) 55-year reanalysis (JRA55; Kobayashi et al. 2015) were used to characterize and detect the shear line. This reanalysis data set has a 1.25° × 1.25° horizontal grid resolution.

Daily rainfall records from 11 weather stations (**Figure 1**) provided by the Department of Science and Technology—Philippine Atmospheric, Geophysical, and Astronomical Services Administration (henceforth, simply referred to as PAGASA) for the period 1979–2022 were used to quantify the contribution of shear lines to extreme rainfall events along the eastern coast of the Philippines. The extreme rainfall days were obtained per month from November to March and were defined as the days that exceeded the 95th percentile rainfall amount per month.

3. Development of a Shear Line Detection Algorithm

3.1. The January 16, 2017 extreme rainfall event as a case study

On January 16, 2017, an extreme rainfall event occurred in eastern Philippines, which was triggered by a shear line that interacted with a low-level cyclonic vortex over Mindanao (Olaguera et al. 2021a). **Figure 2** shows the large-scale circulation features at 06 UTC on January 16, 2017. The confluence zone stretching southwestward from the *col* (indicated by the + mark at approximately 152.5°E, 17°N) and a low-level vortex located over Mindanao can be seen in **Figures 2a** and **2b**. A clear contrast in wind speeds can be seen across the confluence zone, with stronger meridional (V_{925hPa}; **Figure 2a**) and zonal (U_{925hPa}; **Figure 2b**) wind speeds located to the north of the confluence zone. The maximum V_{925hPa} and U_{925hPa} to the north of the confluence zone

are –12.4 m s^{-1} and –16.2 m s^{-1}, respectively. The zero contour line of the V_{925hPa} lies near the confluence zone. The equivalent daily wind fields from January 15 to 17 are shown in **Figures S5.** The interaction of the shear line and the low-level vortex appear short-lived. The low-level vortex can be depicted on January 15 and 16 over Mindanao. By January 17, this vortex weakened, and northeasterly winds are dominant over the entire Philippines.

The lower boundary of the negative meridional gradient of equivalent potential temperature ($\frac{\partial \theta e_{925hPa}}{\partial y}$) can be used to depict the boundary of air masses. However, the zero and negative one contour lines of this variable are spread across the domain (**Figure 2c**). In addition, shear lines may also be located ahead of a cold front (https://rammb2.cira.colostate.edu/wp-content/uploads/2024/07/Fronts_and_Shearlines_Caribbean_20201209.pdf). Thus, this variable alone may not be enough for depicting the location of the shear line but may be useful for depicting the boundary of the cold front (Olaguera and Matsumoto 2020). High values of the boundary layer moisture flux convergence (*VIMFC*; integrated from 1000 hPa to 925 hPa) can also be seen along the confluence zone (**Figure 2d**), another characteristic of the shear line (Olaguera et al. 2021a, b; Reyes et al. 2024). This parameter can be used as a proxy for rainfall, since areas with high *VIMFC* experience high amounts of rainfall (Olaguera et al. 2021a, b).

A shear line is characterized by an abrupt change in the horizontal wind component parallel to it. To determine the regions with such abrupt changes in the horizontal wind component, we first look at the spatial distribution of the gradients of the zonal and meridional winds at 925 hPa (**Figure 3**). Positive $\frac{\partial U_{925hPa}}{\partial x}$, $\frac{\partial V_{925hPa}}{\partial x}$, and negative $\frac{\partial U_{925hPa}}{\partial y}$, $\frac{\partial V_{925hPa}}{\partial y}$ can be seen along the confluence zone, suggesting that it is characterized by strong zonal and meridional gradients. This is consistent with the fact that the winds shift direction and speed along the confluence zone.

Magnitude-wise, we can see the gradients of the U and V winds in the y-direction (i.e., $\frac{\partial U_{925hPa}}{\partial y}$ and $\frac{\partial V_{925hPa}}{\partial y}$) are more defined and continuous compared to those along the x-direction. Also, notice the negative $\frac{\partial U_{925hPa}}{\partial y}$ stretching from western Mindanao to the west of the Philippines (~12°N) without a clear confluence in this region. According to Reyes et al. (2024), a shear line is identified by an area of directional wind confluence at the equatorward side of the *col* area where the front terminates. In **Figure 3**, the *col* is located approximately at 152.5°E, 17°N. Hence, that diagonal region $\frac{\partial U_{925hPa}}{\partial y}$ to the west of the Philippines should no longer be part of the shear line. In addition, the $\frac{\partial V_{925hPa}}{\partial x}$ is positive and $\frac{\partial U_{925hPa}}{\partial y}$ is negative. This indicates that the relative vorticity (RV_{925hPa}) along the shear line is also positive (**Figure 2e**), which could be a result of the strong stretching term in the vorticity budget equation.

Are negative $\frac{\partial V_{925hPa}}{\partial y}$, positive *VIMFC* and positive RV_{925hPa} common features along the shear lines? To answer this question, we looked at other cases triggered by a shear line, as shown in **Supplement 1 (Figures S2 to S5)**. The list of dates was obtained from the website of the Disaster Response Operations Management, Information and Communication, a division of the Disaster Response Management Bureau in the Philippines (https://dromic.dswd.gov.ph; last accessed April 28, 2025). This agency collects information on places and people affected by different hazards affecting the Philippines, including the shear line. The extreme event on January 13, 2009, which was discussed in Faustino-Eslava et al. (2011) was also included. In this event, the rainfall peaked on January 13, 2009, in some PAGASA stations in Mindanao. Negative $\frac{\partial V_{925hPa}}{\partial y}$ and positive *VIMFC* can be seen in all cases, like those in the January 2017 case. Also, notice the low-level cyclonic circulation to the south and near the western edge of the negative $\frac{\partial V_{925hPa}}{\partial y}$. This low-level

cyclonic circulation could facilitate the enhancement of the moist easterly/southwesterly winds that are interacting with the cold northerly/northeasterly winds. Thus, the above conditions seem to be common features of shear lines. The *VIMFC* and RV_{925hPa} are also distinguishable along the region of negative $\frac{\partial V_{925hPa}}{\partial y}$, suggesting that these are key parameters in detecting the shear line.

3.2. Detecting the shear line

The shear line is detected through the following steps, which is also shown in **Figure S6**. For illustrative purposes, we demonstrate the detection of the shear line at 06 UTC on January 16, 2017.

(a) First, the candidate points per longitude are identified within 115–135°E, 5–22°N. This domain is similar to the Philippine Area of Responsibility but with a small latitudinal extent (i.e., 22°N instead of 25°N). This domain was used to filter out the shear lines that are not directly influencing the Philippines, such as those near Taiwan between 22–25°N. The candidate points are the latitude coordinates with at least $\frac{\partial V_{925hPa}}{\partial y} \leq -1 \times 10^{-5}$ s^{-1}, $VIMFC > 0$, and $RV_{925hPa} > 0$. The *VIMFC* condition was added since we are only interested in rainfall inducing shear lines. The $\frac{\partial V_{925hPa}}{\partial y}$ threshold value was determined by trial and error. A sensitivity test of the results to this parameter is discussed in **Supplement 2.**

(b) Second, the candidate grid points were grouped using a simple clustering by proximity (i.e., Single-Linkage clustering; Martinez and Martinez, 2005). A justification for using this clustering methodology is discussed in **Supplement 2**. Each coordinate is assigned to an initial empty cluster. Then, their distances from one another are computed using the Haversine formula. When the distance of one coordinate from another is less than 2.5° (2

grid points or ~277.5 km for the JRA55 data set), they are assigned to the same cluster. Otherwise, a new cluster is created, and the distances are recomputed. This procedure is looped through all the candidate coordinates. Finally, the clusters with at least 10 degrees longitude in extent are obtained. This will filter out the short shear lines.

(c) Third, the clusters in (b) that are at least 555 km (5°) from the zero contour line of the 925 hPa meridional wind are extracted. This was added to the condition since the shear line is close to this zero contour line, as shown in Figure 2a.

(d) A 1-dimensional Gaussian filter is then applied to both filtered latitude and longitude points in (c). We used a value of 3 for the standard deviation of the Gaussian function, which controls the smoothing of the shear line.

The detected shearline at 925 hPa can be seen in **Figure 4a**. It is located to the north of Mindanao, extending from (122°E, 10°N) to (134°E, 12°N). The detection algorithm using the same criteria above was applied at 1000 hPa, 975 hPa, 950 hPa, and 850 hPa pressure levels to determine whether this system is best depicted at 925 hPa. The shearline seems to be detectable at these four pressure levels. At 1000 hPa (**Figure 4b**), 975 hPa (**Figure 4b**), and 950 hPa (**Figure 4c**) the shear lines are less smooth than those at 925 hPa and 850 hPa. The $\frac{\partial V_{925hPa}}{\partial y}$ appears strongest at 925 hPa level.

All seven shear line cases shown in **Supplement 1** were detected by the algorithm (**Figure S5**), indicating its good performance. We also compared detected shear line days with those obtained by Reyes et al. (2024; personal communication). About 72% of their detected shear line

194 days matched with that obtained in this study. More discussions on this discrepancy are provided
195 in **Supplement 2**.

196 4. Climatology of detected shear lines and their contribution to extreme rainfall

197 **Figure 5** shows the density map of the detected shear lines from 1979 to 2022. A total of 2,432
198 days with shear lines (i.e., shear line days) were detected, which is about 36.5% of the total days
199 (6,504 days). About 51%, 46%, 38%, 30%, and 22% of the total days in November, December,
200 January, February, and March, have shear lines that are within the domain in **Figure 5**,
201 respectively. Spatially, shear lines appear more frequently to the north of 10°N, particularly in
202 November and December, compared to its southern region. We also counted the "shear line
203 events," where shear lines were tracked across space and time, with a unique ID assigned to
204 consecutive occurrence in time at least 555 km (5°) apart (i.e., this identifies the same shear line that is
205 propagating in space in time and assigned with only one unique ID). This yielded 17, 14, 13, 11, and 11
206 shear line events per month for November, December, January, February, and March, respectively.

207 The contribution of shear lines to extreme rainfall along the eastern coast of the Philippines was
208 examined. A shear line is considered to have contributed to the extreme rainfall days in a station
209 if it is located within a 150 km radius centered at the station location. The percentage contribution
210 is the ratio of the extreme rainfall days associated with the shear lines to the total number of
211 extreme rainfall days, expressed as a percentage in each station (**Table 1**). Overall, shear lines
212 account for up to 20% of the total number of extreme rainfall days. Specifically, stations in Luzon
213 (Casiguran, Infanta, Daet, Alabat, Tayabas, Virac, and Legazpi City) experience higher extreme
214 rainfall days triggered by the shear line during November and December, while those in the

Visayas (Catarman and Guiuan) and Mindanao (Surigao City) experience more extreme rainfall days from November to January. A higher percentage of extreme days associated with shear lines occurs in January at Hinatuan station in Mindanao, which is the southernmost station along the eastern coast and the wettest location in the Philippines (Olaguera and Manalo, 2024a).

5. Summary and Conclusions

Shear lines can trigger heavy to extreme rainfall events during the northeast monsoon season in the Philippines. Therefore, it is important to monitor its location for improved forecasting and disaster mitigation. Using the January 16, 2017 heavy rainfall event as a case study, the results show that the $\frac{\partial V_{925hPa}}{\partial y}$ is a parameter that can be used in determining the location of the shear line. A clear shear zone characterized by a strong negative $\frac{\partial V_{925hPa}}{\partial y}$ was identified. Since enhanced convergence can be seen along the shear line, this parameter may be combined with the *VIMFC*. In addition, the RV_{925hPa} appears to be strong along the region of negative $\frac{\partial V_{925hPa}}{\partial y}$ suggesting that this is another parameter that could be used for the detection of the shear line. We also noticed that the shear line lies near the zero contour line of the 925 hPa meridional winds (V_{925hPa}).

A shear line detection algorithm was developed using a threshold value of $\frac{\partial V_{925hPa}}{\partial y} \leq 1 \times 10^{-5}\ s^{-1}$, positive *VIMFC*, positive RV_{925hPa}, and a length condition (i.e., the shear line must be at least 10 degrees longitude in length). The detected candidate grid points must also be close to the zero contour line of the 925 hPa (i.e., at least 555 km (5°) from the $V_{925hPa} = 0$ contour line).

The shear line is best depicted at 925 hPa and 850 hPa. It is detectable at other pressure levels but appears less smooth. When the detection algorithm was applied to climatological data, shear lines

were found to occur more frequently between November and January, with a peak in November. This coincides with the onset of the northeast monsoon season along the eastern coast of the Philippines (Kubota et al. 2017; Matsumoto et al. 2020). They are also more frequent to the north of 10°N (i.e., in Visayas and Luzon), particularly in December, compared to the south (i.e., in Mindanao). Furthermore, shear lines account for no more than 20% of the total incidence of extreme rainfall days observed at eastern coastal stations in the Philippines. To test the effectiveness of the detection algorithm, we demonstrate that it can be applied to a forecast of a heavy rainfall event in the Philippines, as explained in **Supplement 3**. The algorithm was able to depict the forecast position of the shear line that induced heavy rainfall along the eastern coast of the Philippines on January 9, 2025.

There are still some areas to explore in future work. As mentioned in the discussion, only 72% of the shear line days analyzed by Reyes et al. (2024) were detected by the developed detection algorithm. Perhaps additional criteria such as the gradient of the divergence field (e.g., Berry and Reeder, 2014) may be added to improve the detection of shear lines. The performance of the detection algorithm on high-resolution datasets also needs to be tested further in future works. As demonstrated in **Supplement 3**, the smoothing function was adjusted since the wind fields are noisier in the high-resolution data. The precise evolution of the shear line and its interaction with other systems were not examined in detail in this study. Furthermore, not all shear lines may cause heavy to extreme rainfall in the Philippines. The conditions responsible for shear line-induced heavy to extreme rainfall warrant further investigation.

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Supplements

Supplement 1

Figure S1. Spatial distribution of daily mean zonal winds at 925 hPa (U_{925hPa}; shades; m s^{-1}), the zero U_{925hPa} contour line (blue contour line), and winds at 925 hPa ($WINDS_{925hPa}$; streams; m s^{-1}) (a, c, e) from January 15 to 17, 2017. Fig. S1 (b, d, and f) are for the meridional winds at 925 hPa (V_{925hPa}).

Figure S2. Spatial distribution of the 925 hPa meridional gradient of meridional winds ($\frac{\partial V_{925hPa}}{\partial y}$; $\times 10^{-5}$ s^{-1}). The streamlines are the winds at 925 hPa.

Figure S3. As in Fig. S2, but for the boundary layer moisture flux convergence (integrated from 1000 hPa to 925 hPa; VIMFC; $\times 10^{-4}$ kg m^{-2} s^{-1}).

Figure S4. As in Fig. S2, but for the 925 hPa relative vorticity (RV_{925hPa}; $\times 10^{-5}$ s^{-1}).

Figure S5. The detected shearlines in the dates shown in Figs. S2, S3, and S4.

Supplement 2 Additional information on the detection algorithm.

Figure S6. The methodology for estimating the location of the shear line.

Figure S7. Spatial distribution of (a) $\frac{\partial V_{925hPa}}{\partial y}$ ($\times 10^{-5}$ s^{-1}), (b) VIMFC ($\times 10^{-4}$ kg m^{-2} s^{-1}), and (c) RV_{925hPa} ($\times 10^{-5}$ s^{-1}) on January 25, 2022, 2 PM LST. The green dot-dashed line is the shearline detected without the RV_{925hPa} criteria. The dark blue line in (a) is the zero contour line of the meridional wind at 925 hPa.

Figure S8. Spatial distribution of 6-hourly accumulated rainfall from the gauge-adjusted Global Satellite Mapping of Precipitation version 8 (GSMaP v8; Kubota et al. 2020). The buffer lines represent the radius from the coastline, ranging from 50 to 300 km, at 50 km intervals. The 150 km buffer zone is represented by the thick solid line.

Figure S9. The detected shear lines (green dot-dashed line) at 06 UTC of January 16, 2017, using different values of $\frac{\partial V_{925hPa}}{\partial y}$.

Supplement 3 Application of the detection algorithm on a heavy rainfall forecast.

Figure S10. Spatial maps of forecast (a) $\frac{\partial V_{925hPa}}{\partial y}$ (shades; $\times 10^{-5}$ s^{-1}), (b) Precipitation rate (mm hr^{-1}), and (c) boundary moisture flux convergence (*VIMFC*; integrated from 1000 hPa to 925 hPa; $\times 10^{-4}$ kg m^{-2} s^{-1}) at 11 PM of January 9, 2025. The forecast was initialized at 8 AM (00 UTC) on January 7, 2025. The streamlines are the winds at 925 hPa (m s^{-1}). The light green dot-dashed line indicates the location of the shear line. The black areas in (a) are elevations above 760 m (925 hPa).

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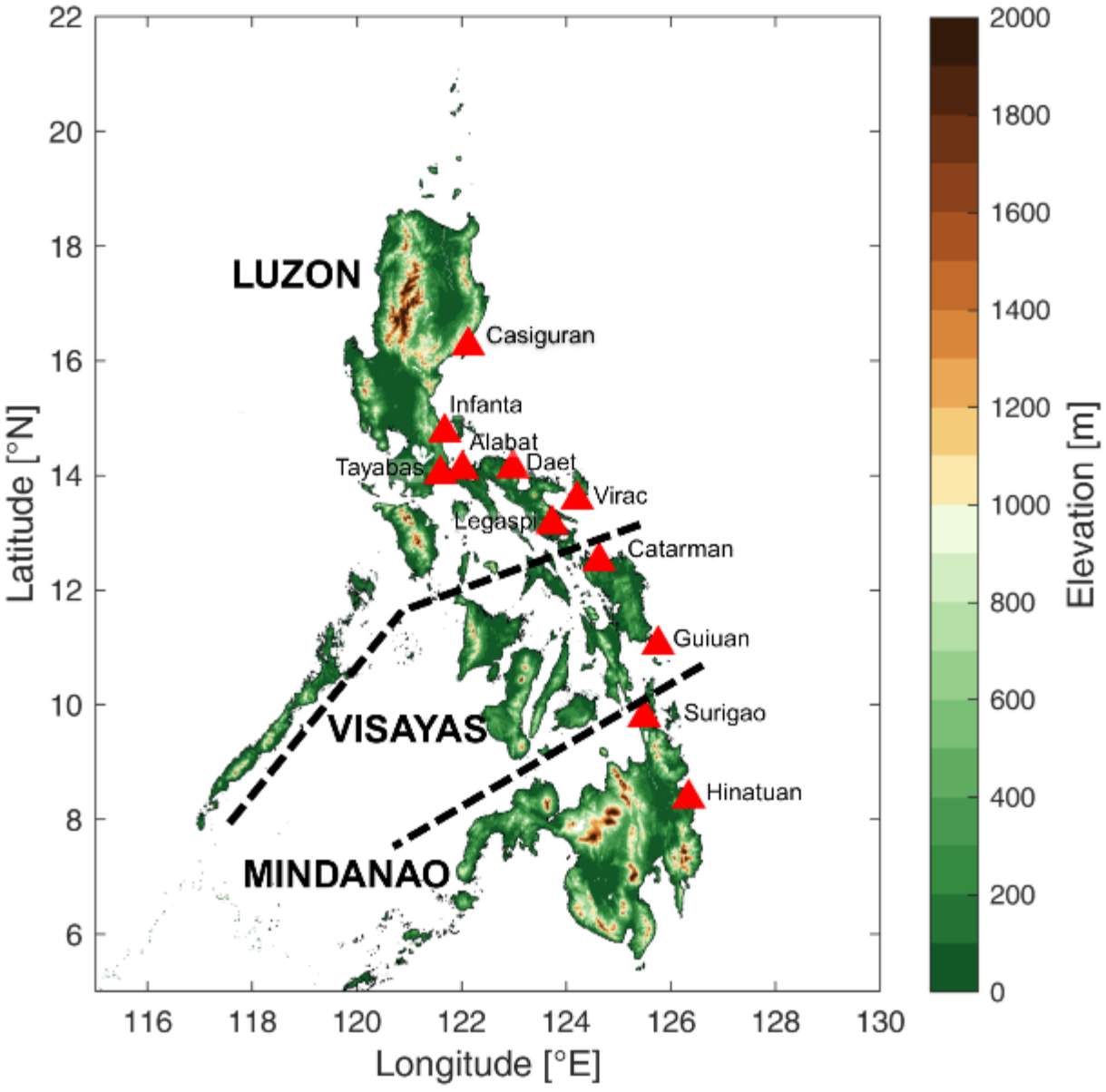


Figure 1. Location of the 11 PAGASA stations used in this study. The extent of the domain shown here was used in detecting the shear lines. The broken lines divide the Philippines into three major island groups: Luzon, Visayas, and Mindanao.

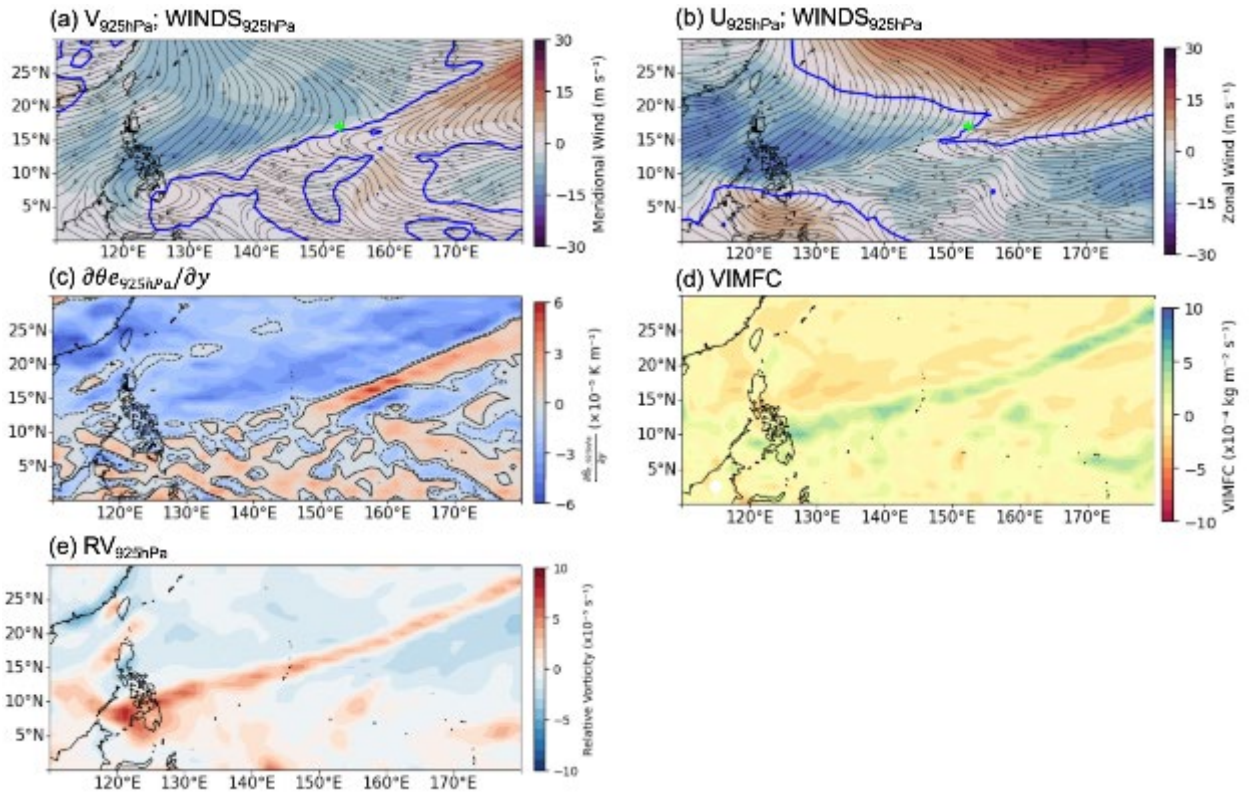


Figure 2. Spatial distribution of the 925 hPa (a) meridional (V_{925hPa}; shades; m s^{-1}) and (b) zonal winds (U_{925hPa}; shades; m s^{-1}), (c) meridional gradient of equivalent potential temperature ($\frac{\partial \theta e_{925hPa}}{\partial y}$; × 10^{-5} K m^{-1}), (d) boundary layer moisture flux convergence (integrated from 1000 hPa to 925 hPa; *VIMFC*; ×10^{-4} kg m^{-2} s^{-1}), and (e) 925 hPa relative vorticity (RV_{925hPa}; ×10^{-5} s^{-1}) at 06 UTC of January 16, 2017. The streamlines in (a) and (b) are the winds at 925 hPa. The thick blue lines are the zero value of the meridional and zonal winds. The solid (dashed) black line in (c) indicates the zero (negative one) contour line of $\frac{\partial \theta e_{925hPa}}{\partial y}$. The green "+" in (a) and (b) indicates the approximate location of the *col*.

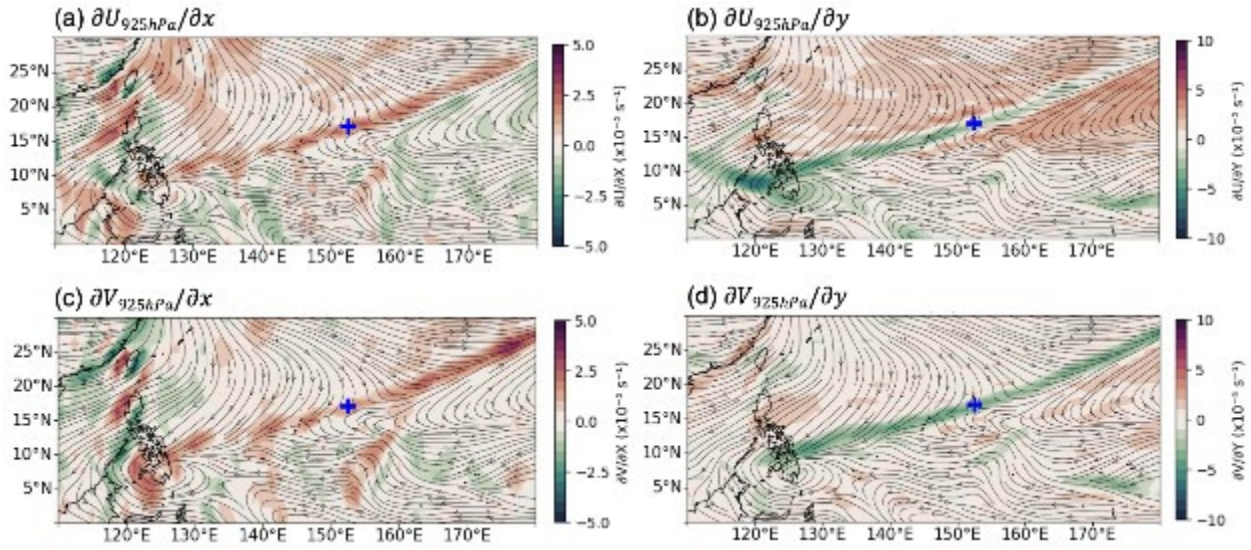


Figure 3. Spatial maps of (a) $\frac{\partial U_{925hPa}}{\partial x}$, (b) $\frac{\partial U_{925hPa}}{\partial y}$, (c) $\frac{\partial V_{925hPa}}{\partial x}$, and (d) $\frac{\partial V_{925hPa}}{\partial y}$ at 06 UTC of January 16, 2017. The unit is in $\cdot 10^{-5}$ s^{-1}. The streamlines are the 925 hPa winds. The blue "+" mark indicates the approximate location of the *col*.

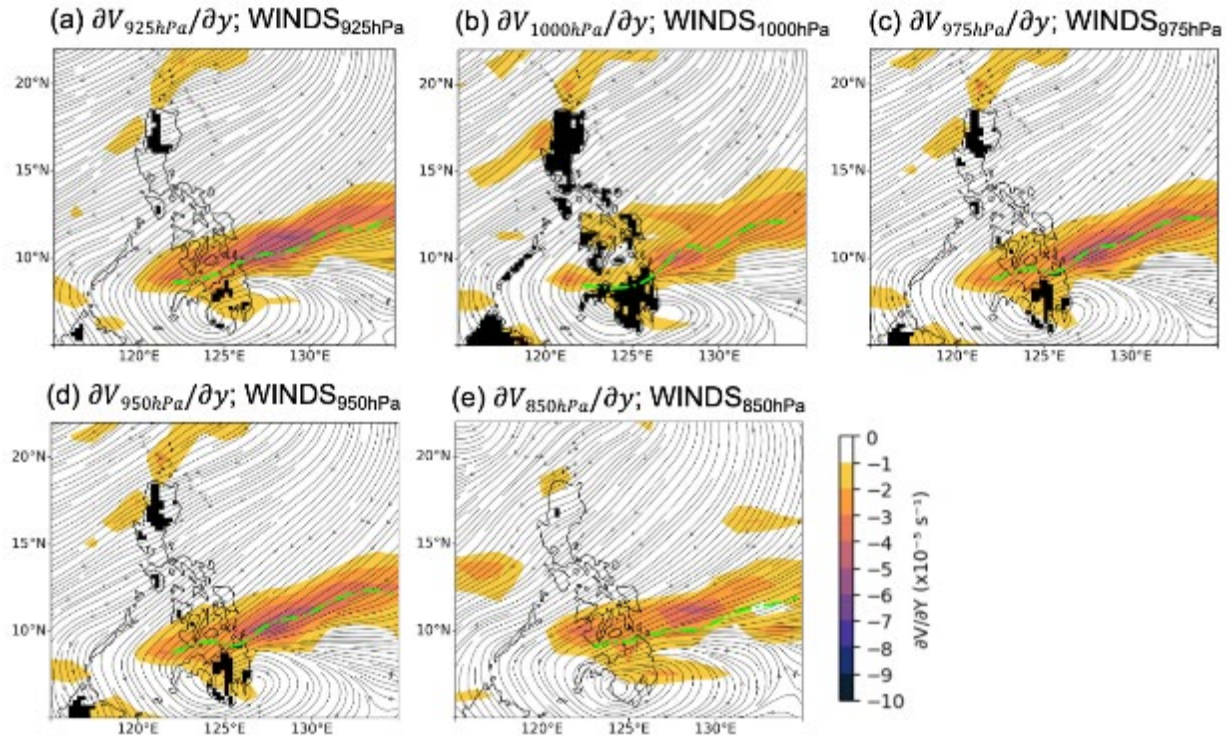


Figure 4. The detected shear line (green dot-dashed line) at (a) 925 hPa, (b) 1000 hPa, (c) 975 hPa, (d) 950 hPa, and (e) 850 hPa at 06 UTC of January 16, 2017. The streamlines are the winds at each pressure level. The shaded areas are the $\frac{\partial V_{925hPa}}{\partial y} \leq -1 \times 10^{-5}$ s^{-1}. The black areas are the masked elevations at each pressure level.

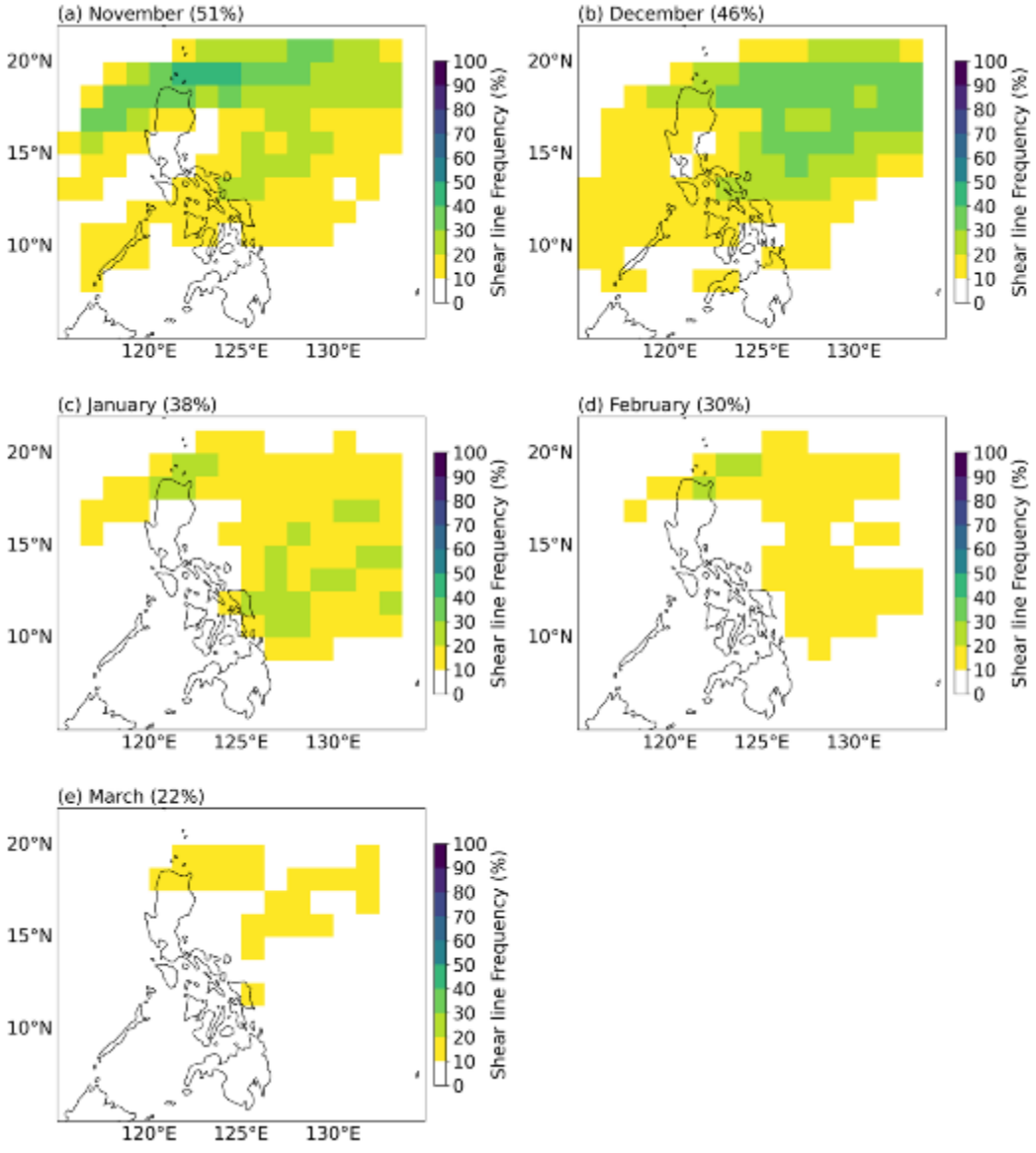


Figure 5. Spatial distribution of detected shear lines frequency (i.e., here defined as the total number of unique days with shear lines in a month divided by the total number of days in a month expressed as percentage) at 925 hPa from November to March. The frequency of shear lines is accumulated at each 1.25°×1.25° grid boxes between 115°–135°E,

445 5°–22°N. The percentage of days with unique shear lines per month is
446 enclosed in parentheses. Grid boxes with no values are colored white.
447
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449 **List of Tables**

450

451 **Table 1.** Percentage contribution of shear lines to the 95th percentile extremes across 11 PAGASA
452 stations along the eastern coast of the Philippines. The stations are arranged from north to south.

Station/Month	January	February	March	November	December
Casiguran	0.0	1.2	0.6	10.4	13.4
Infanta	0.5	1.4	0.5	8.4	12.1
Alabat	1.8	1.2	1.8	9.4	12.9
Tayabas	0.0	1.7	0.6	12.8	8.1
Daet	4.5	2.0	3.5	12.4	16.9
Virac (Synop)	9.5	3.6	1.2	14.2	19.5
Legazpi City	5.5	5.0	3.0	9.0	17.1
Catarman	16.4	5.6	5.6	14.6	19.2
Guiuan	16.4	10.5	7.3	8.2	12.3
Surigao City	17.5	11.2	6.3	4.9	11.7
Hinatuan	12.3	6.1	2.6	0.9	6.6

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Statements and Declarations

Competing Interests

The authors declare that they have no competing interests.

Declaration of Interests

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

SUPPLEMENT 1

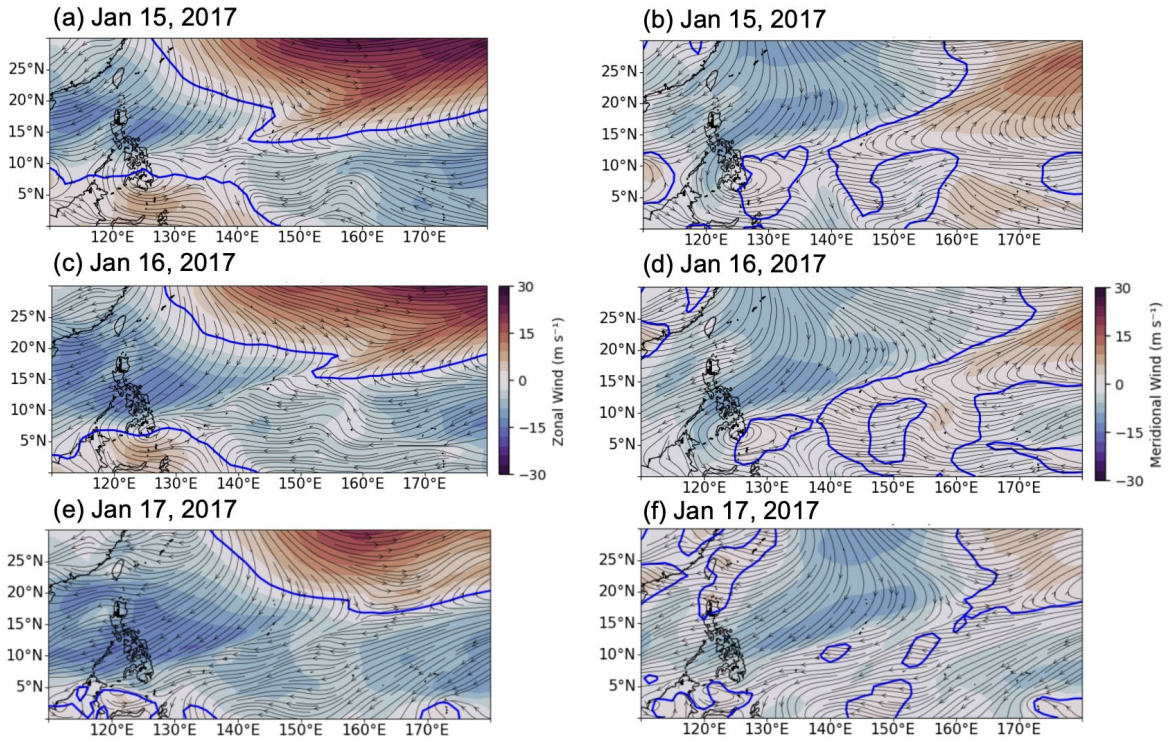


Figure S1. Spatial distribution of daily mean zonal winds at 925 hPa (U_{925hPa}; shades; m s^{-1}), the zero U_{925hPa} contour line (blue contour line), and winds at 925 hPa ($WINDS_{925hPa}$; streams; m s^{-1}) (a, c, e) from January 15 to 17, 2017. Fig. S1 (b, d, and f) are for the meridional winds at 925 hPa (V_{925hPa}).

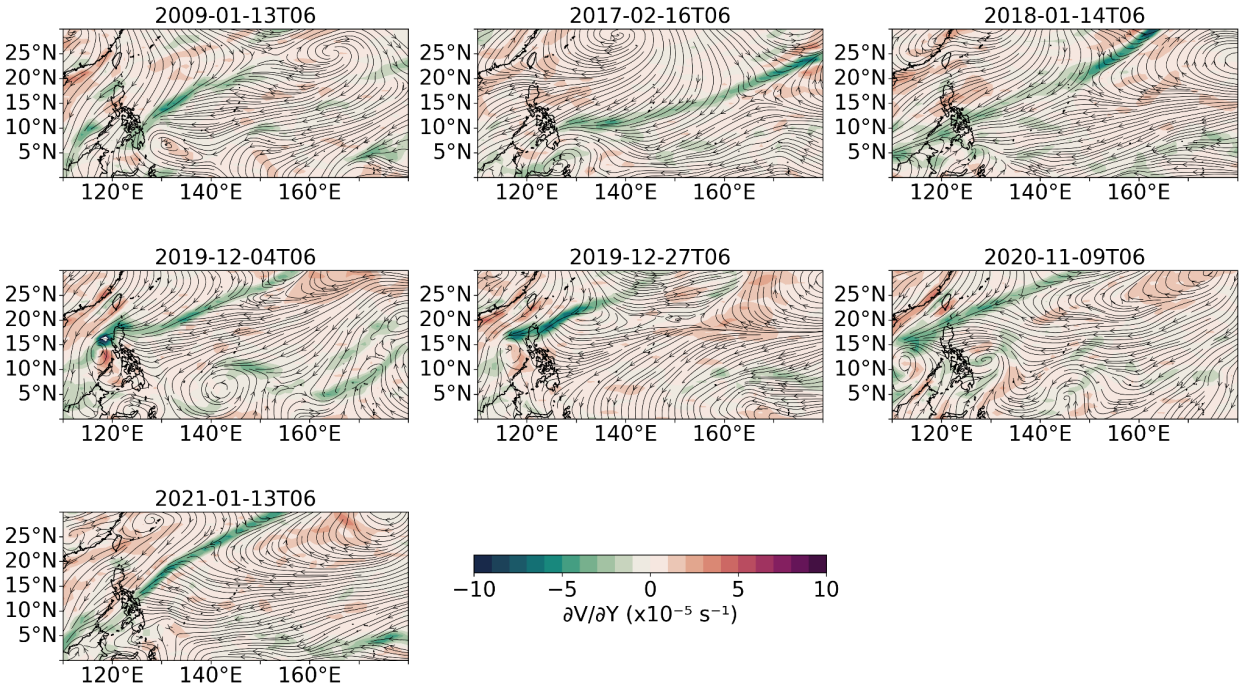


Figure S2. Spatial distribution of the 925 hPa meridional gradient of meridional winds ($\frac{\partial V_{925hPa}}{\partial y}$; $\times 10^{-5}$ s^{-1}). The streamlines are the winds at 925 hPa.

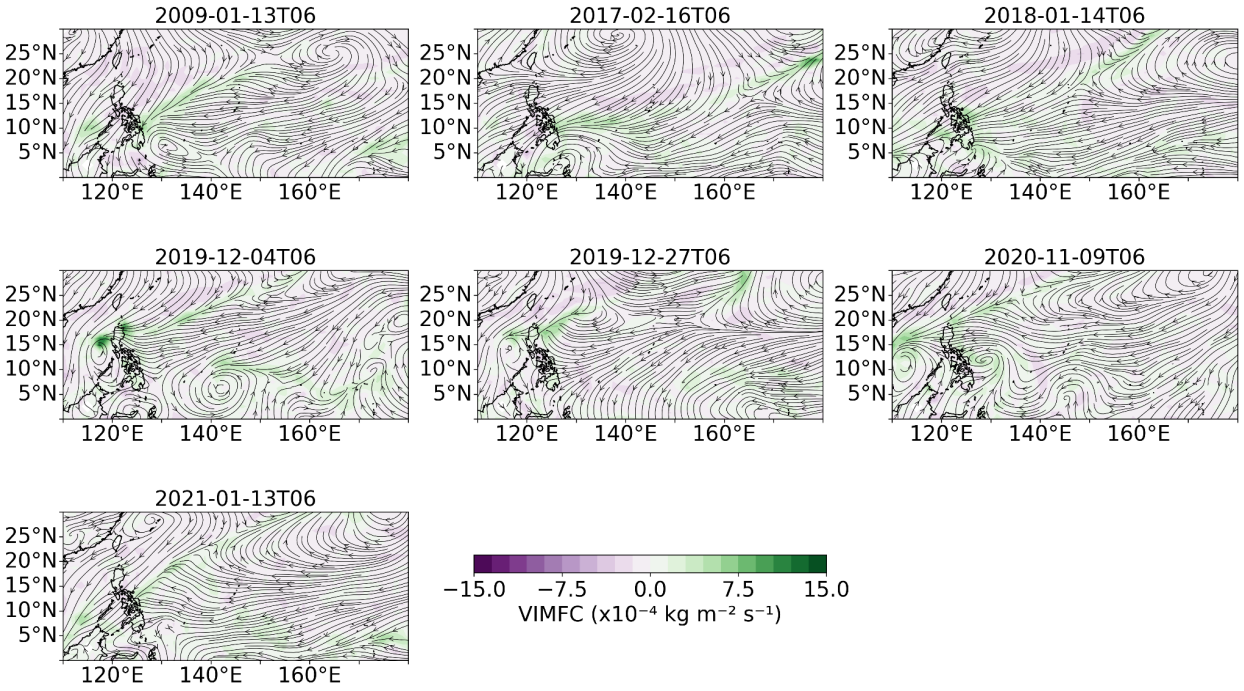


Figure S3. As in Fig. S2, but for the boundary layer moisture flux convergence (integrated from 1000 hPa to 925 hPa; VIMFC; $\times 10^{-4}$ kg m^{-2} s^{-1}).

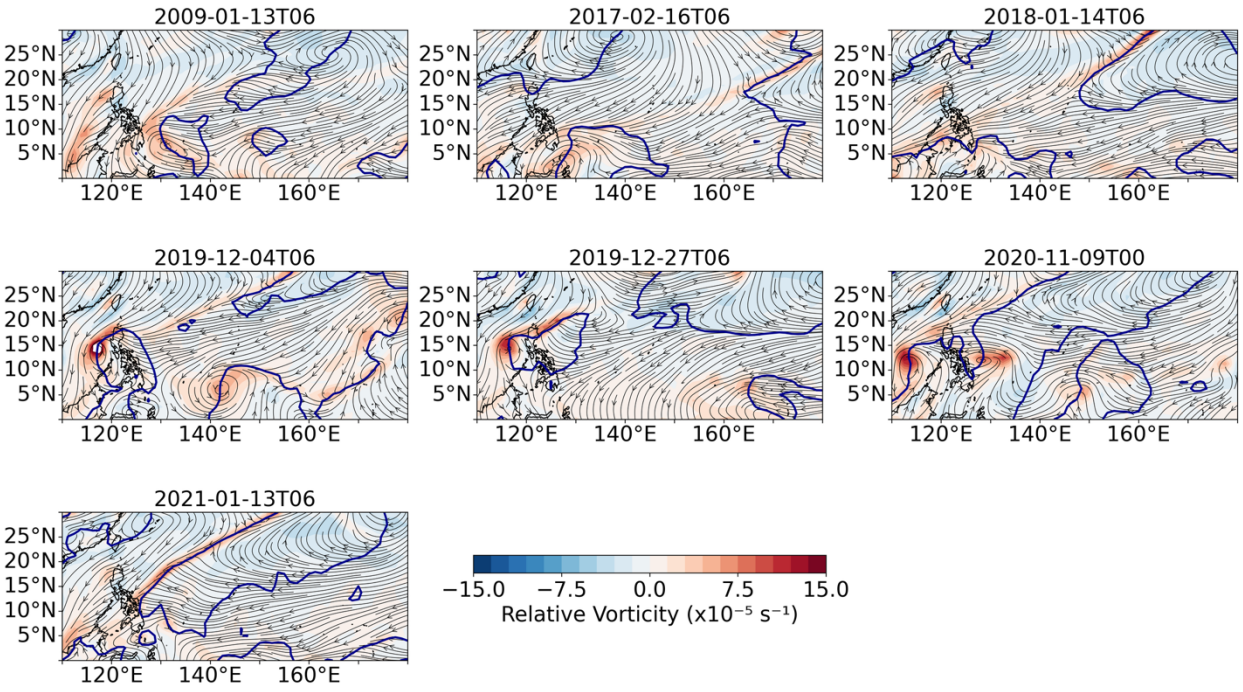


Figure S4. As in Fig. S2, but for the 925 hPa relative vorticity (RV_{925hPa}; $\times 10^{-5}$ s^{-1}). The dark blue line is the zero contour of the 925 hPa meridional wind.

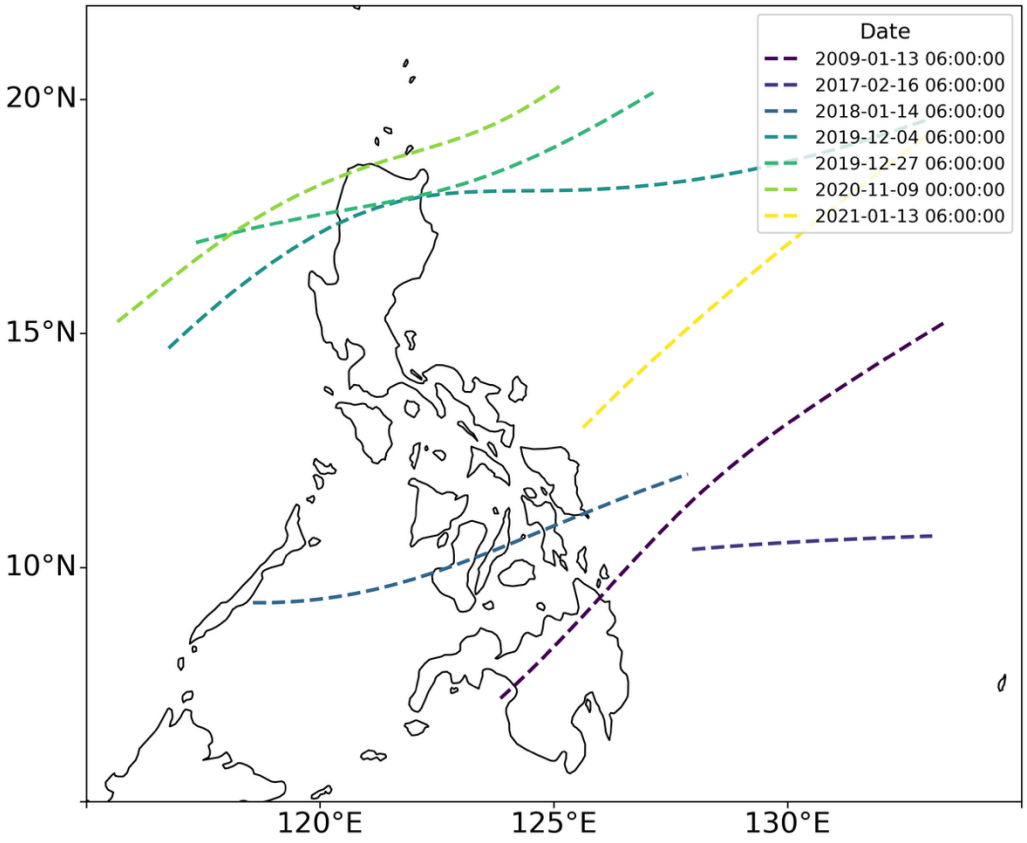


Figure S5. The detected shearlines in the dates shown in Figs. S2, S3, and S4.

SUPPLEMENT 2

Additional information on the detection algorithm

In this section, we discuss the methodology and the sensitivity tests that we conducted in developing the detection algorithm. Determining the appropriate radius for estimating the extreme rainfall contribution of shear lines is also discussed.

S2.1. The flowchart of the shear line detection algorithm

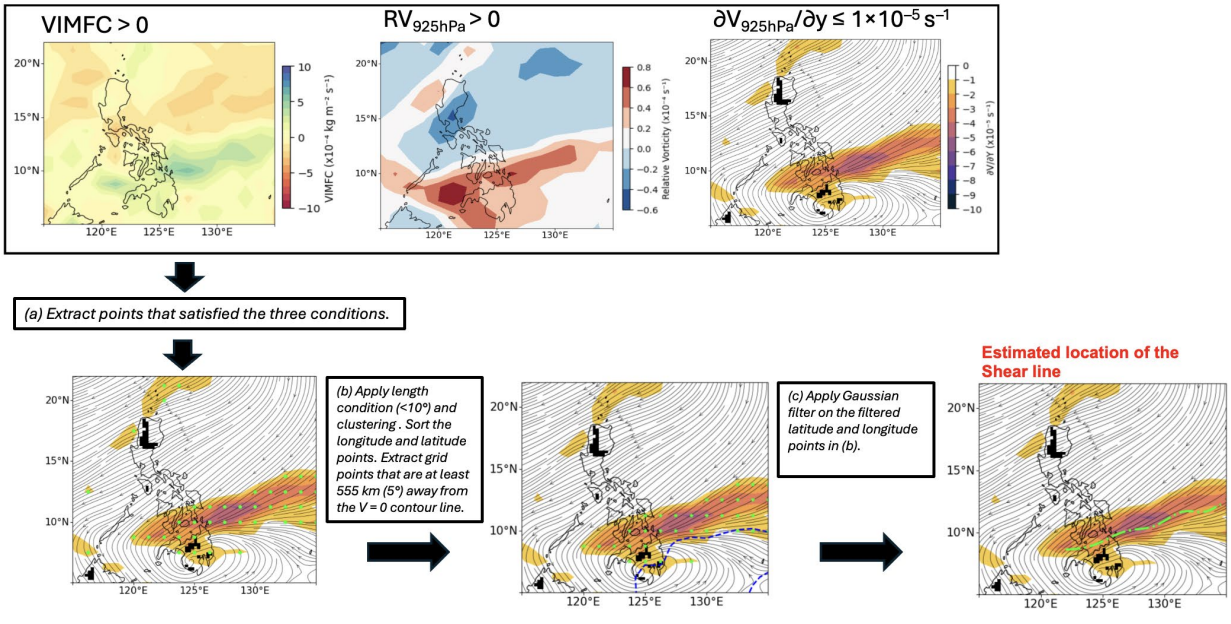


Figure S6. The methodology for estimating the location of the shear line.

As outlined in Section 3 in the main text, the shear line was detected through the following steps, as shown in **Figure S6**. The 06 UTC on January 16, 2017 was used as a test case.

Step 1: First, the candidate points per longitude are identified within 115–135°E, 5–22°N. These candidate points are the latitude coordinates with at least $\frac{\partial V_{925hPa}}{\partial y} \leq -1 \times 10^{-5}$ s^{-1}, $VIMFC > 0$, and $RV_{925hPa} > 0$. These candidate grid points were then grouped using a simple clustering by proximity (i.e., Single-Linkage clustering). Then, their distances from one another are computed using the Haversine formula. When the distance of one coordinate from another is less than 2.5° (2 grid points or ~277.5 km), they are assigned to the same cluster. Otherwise, a new cluster is created, and the distances are recomputed. This procedure is looped through all the candidate coordinates. Finally, the shear lines with at least 10 degrees longitude are obtained.

Step 2: The clusters in (b) that are close to the zero contour line of the 925 hPa meridional winds are extracted.

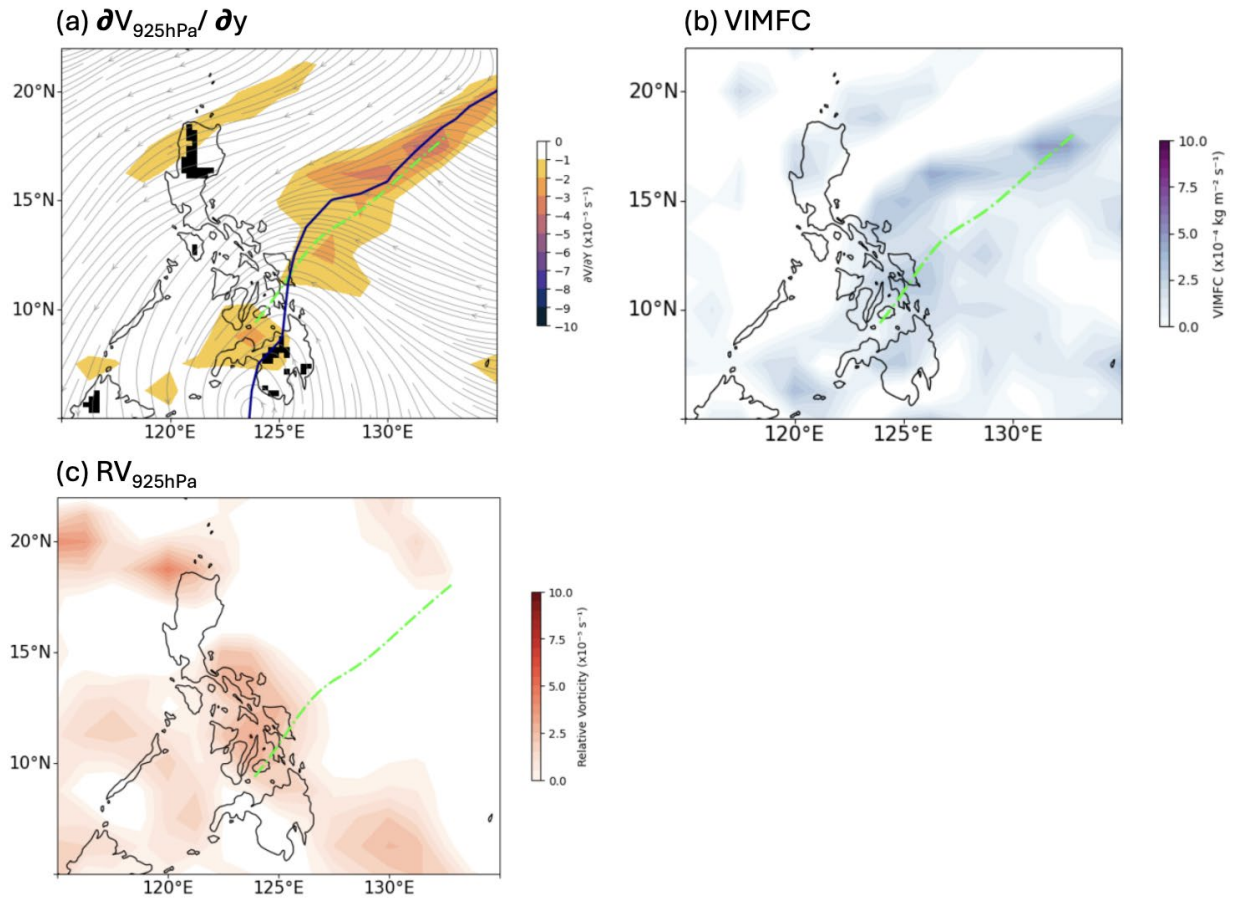
Step 3: A 1-dimensional Gaussian filter was then applied to filtered latitude and longitude points obtained in Step 2.

S2.2. The choice of the clustering technique

We tested another clustering method, the Density-Based Spatial Clustering of Applications with Noise (DBSCAN; Ester et al., 1996; Schubert et al., 2017), in addition to the Single-Linkage Clustering (SLC). This method is used for spatial clustering and can cluster points with different shapes and densities. However, two key input parameters require tuning when using this method: the radius of the neighborhood (eps parameter) and the minimum number of points required to form a dense region (*minpts*). For the January 16, 2017, case, we found that *eps* = 2 and *minpts* = 2 are enough to detect the shear line on this date. These values may vary per year and need to be estimated when applied to climatological data. This will not be practical for forecasting

77 applications. The *k*-means clustering, on the other hand, which is another popular clustering
78 technique, assumes that the shape of the clusters is spherical, which is not appropriate for detecting
79 shear lines. The SLC used in this study does not require the eps and *minpts* conditions, unlike
80 DBSCAN. The SLC method may also be used for clustering elongated or chain-like clusters,
81 whose density may not be enough to be recognized by DBSCAN.
82
83 **S2.3. Inclusion of the VIMFC in the detection criteria**
84
85 We used the VIMFC as a proxy for rainfall, focusing specifically on shear lines associated with
86 heavy to extreme precipitation. It can also be seen in the seven cases in Supplement 1, and the case
87 study on January 16, 2017, the region with $\frac{\partial V_{925hPa}}{\partial y} \leq -1 \times 10^{-5}$ s^{-1} coincides with the region
88 with *VIMFC* > 0. The VIMFC condition is important for filtering the rainfall-inducing shearlines.
89
90 **S2.4. Inclusion of 925 hPa Relative Vorticity (RV_{925hPa}) in the criteria**
91
92 In the January 16, 2017 event, as well as in the seven shear line cases in Supplement 1, a positive
93 relative vorticity at 925 hPa (i.e., $RV_{925hPa} > 0$) can be seen along the shear line. About 3117 unique
94 shear line days are detected by the algorithm that does not incorporate this condition. After
95 including this condition, the number of detected shear lines is reduced to 2432. We show an
96 example of a case on January 25, 2022, 2PM Local Standard Time (LST) that was detected by the
97 algorithm without the RV_{925hPa} condition (**Figure S7**). To the best of our knowledge, there was no
98 recorded shear line warning or event issued on this day. It can be seen that there is no positive
99 RV_{925hPa} at the location of the detected shear line, particularly to the northeast of Visayas. The

100 RV_{925hPa} can be used as an indicator of the low-level convergence that maintains the shear line
101 (Pielke, 2013). Therefore, the $RV_{925hPa} > 0$ eliminates the cases with no low-level convergence
102 along the shear line.



103

104

105 **Figure S7.** Spatial distribution of (a) $\frac{\partial V_{925hPa}}{\partial y}$ ($\times 10^{-5}$ s^{-1}), (b) VIMFC ($\times 10^{-4}$ kg m^{-2} s^{-1}), and (c)
106 RV_{925hPa} ($\times 10^{-5}$ s^{-1}) on January 25, 2022, 2 PM LST. The green dot-dashed line is the shearline
107 detected without the RV_{925hPa} criteria. The dark blue line in (a) is the zero contour line of the
108 meridional wind at 925 hPa.

109 **S2.5. Estimating the radius for extreme rainfall analysis**

The composite map of 6-hourly accumulated rainfall for the days with shear lines from the gauge-adjusted Global Satellite Mapping of Precipitation version 8 (GSMaP v8; Kubota et al., 2020) data set from 2000 to 2022 is shown in **Figure S8**. Buffer zones ranging from 50 to 300 km at 50 km intervals were created from the coastline to determine the optimal radius for assessing the influence of shear lines on extreme rainfall at various stations. It appears that a 150 km buffer zone is enough since the maximum rainfall (~5 mm) along the eastern coast is within this zone.

S1.6. Sensitivity to the $\frac{\partial V_{925hPa}}{\partial y}$ threshold

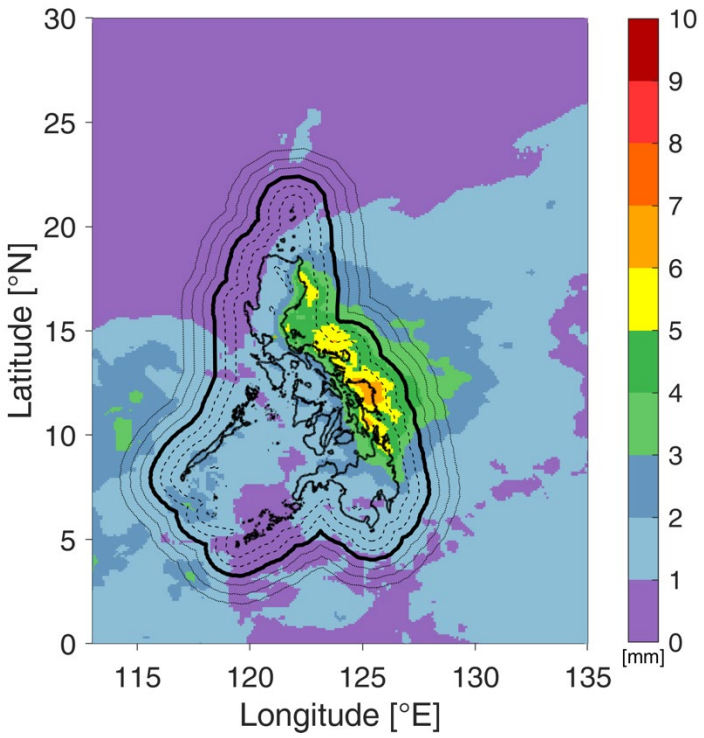


Figure S8. Spatial distribution of 6-hourly accumulated rainfall from the gauge-adjusted Global Satellite Mapping of Precipitation version 8 (GSMaP v8; Kubota et al. 2020). The buffer lines represent the radius from the coastline, ranging from 50 to 300 km, at 50 km intervals. The 150 km buffer zone is represented by the thick solid line.

Figure S9 shows the impact of varying the threshold of the $\frac{\partial V_{925hPa}}{\partial y}$. When the value of this parameter is less than 0, the detected shear line appears wavy. This behavior lessens upon decreasing the value to -1×10^{-5} s^{-1}. A straight line is achieved when this is set to -3×10^{-5} s^{-1}. No shear lines were detected for the -4×10^{-5} s^{-1}. This is because the area with the $<-4 \times 10^{-5}$ s^{-1} is already small (i.e., less than 10°). In this study, we used -1×10^{-5} s^{-1} since it appears to give already a nearly straight line and all the shear lines shown in **Figure S5** were detected.

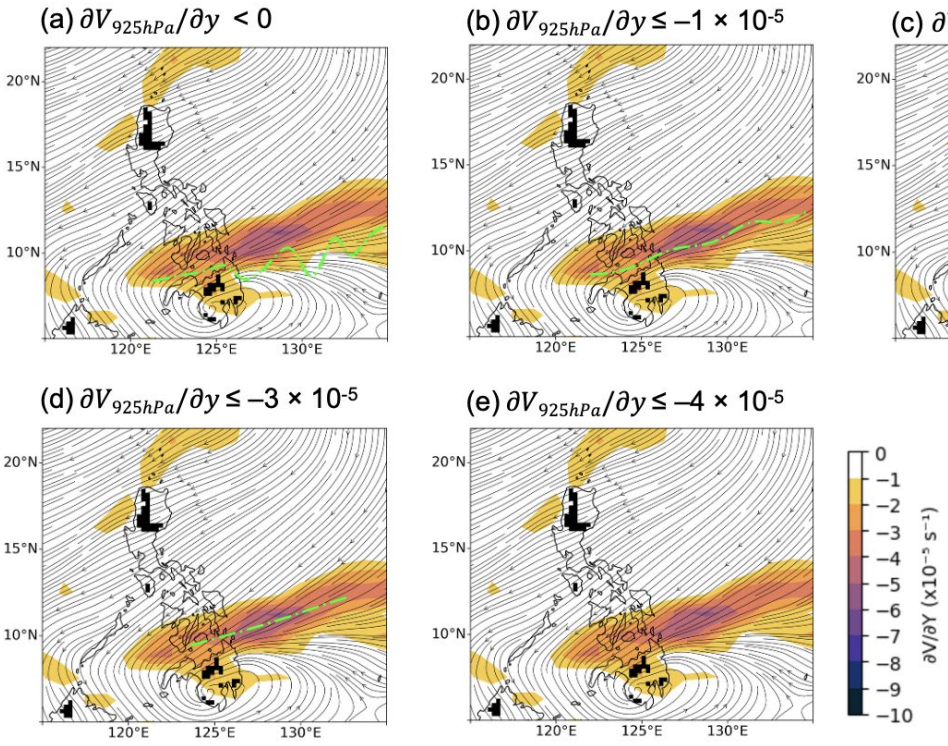


Figure S9. The detected shear lines (green dot-dashed line) at 06 UTC of January 16, 2017, using different values of $\frac{\partial V_{925hPa}}{\partial y}$.

S2.7. Comparison of detected shear line dates with literature

We obtained the dates of the HREs associated with shear lines for the period 2003–2022 from Reyes et al. (2024; personal communication). Their analysis identified a total of 152 HREs using a threshold of 50 mm day^{-1}. They applied principal component analysis to the observed data, extracting 18 principal components, which were subsequently grouped into three clusters. The synoptic conditions based on the fifth-generation European Centre for Medium-Range Weather Forecasts (ERA5) reanalysis data associated with each cluster were then examined. Upon comparison, about 72% of the dates (109 unique shear line dates) they identified matched those found in our analysis. Notably, their study was based on daily data. In contrast, our approach differs significantly. In this study, we first objectively detected shear lines using the 6-hourly JRA55 reanalysis data, before identifying the corresponding dates, which is nearly the reverse of their methodology. Further investigation into why some of the days in Reyes et al. (2024) were not detected by the objective method developed in this study reveals that this is due to the threshold set for the $\frac{\partial V_{925hPa}}{\partial y}$. When we set it to $\frac{\partial V_{925hPa}}{\partial y} < 0$, all dates matched, but some shear lines appeared wavy, indicating that the algorithm is detecting some noise (not shown).

SUPPLEMENT 3

Application of the detection algorithm on a heavy rainfall forecast

In this section, the detection algorithm was applied to higher-resolution forecast data to demonstrate its effectiveness in depicting the shear line. On January 9, 2025, a shear line was reported to be affecting southern Luzon and some parts of eastern Visayas (https://ndrrmc.gov.ph/attachments/article/4281/Progress_Report_No_7_for_the_Effects_of_Shear_Line_in_Bicol_Region.pdf). Flooding incidents were reported in these areas.

We used the forecast data from the National Centers for Environmental Prediction's Global Forecast System (NCEP-GFS; for brevity, simply referred to as GFS) (https://www.nco.ncep.noaa.gov/pmb/products/gfs/). This data set has a resolution of 0.25° × 0.25°. The forecasts are initialized four times a day at 00, 06, 12, and 18 UTC. Data from the operational forecasts are available at hourly intervals, while the archived forecasts are only available at 3-hourly intervals for the first 240 time steps. We used the archived forecasts in this study.

The distance condition in identifying unique shear lines was adjusted to 50 km (2 grid points, same as in the JRA55 data set) and the standard deviation of the Gaussian filter was adjusted to seven in the tracking algorithm. As can be seen in **Figure S10**, there are more areas with $\leq -1 \times 10^{-5}$ s^{-1}; hence, more candidate grid points for the shear lines. The flooding on January 9,

2025, was caused by the shear line extending from the central Philippines to the northeast. High VIMFC can be seen along the shear line (**Figure S10c**). Positive RV_{925hPa} is also distinguishable along the shear line. More importantly, the GFS data was able to capture the high precipitation amounts over eastern Visayas. This suggests that this event is predictable.

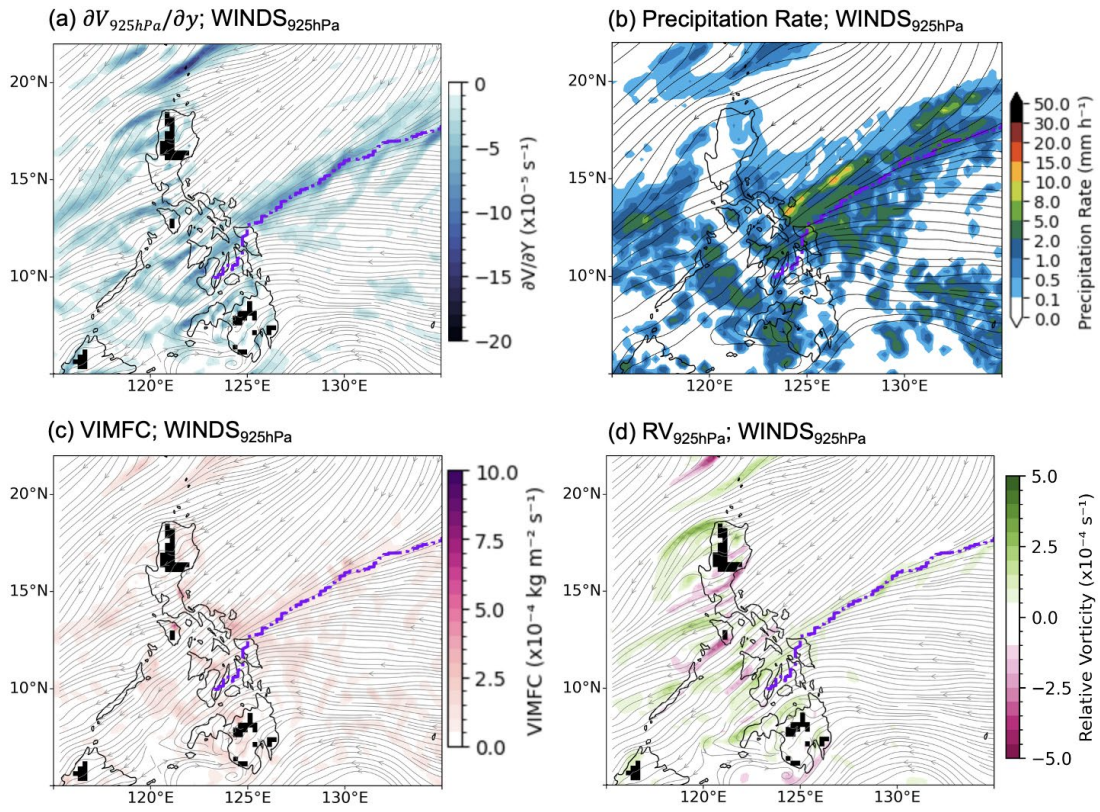


Figure S10. Spatial maps of forecast (a) $\frac{\partial V_{925hPa}}{\partial y}$ (shades; $\times 10^{-5}$ s^{-1}), (b) Precipitation rate (mm hr^{-1}), (c) boundary moisture flux convergence (VIMFC; integrated from 1000 hPa to 925 hPa; $\times 10^{-4}$ kg m^{-2} s^{-1}), and 925 hPa relative vorticity (RV_{925hPa}; $\times 10^{-4}$ s^{-1}) at 11 PM of January 9, 2025. The forecast was initialized at 8 AM (00 UTC) on January 7, 2025. The streamlines are the winds at 925 hPa (m s^{-1}). The light green dot-dashed line indicates the location of the shear line. The black areas in (a) are elevations above 760 m (925 hPa).

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